Data and Signals

Maria Kihl

Book chapters

Forouzan: 1.1, 3, 4.2, 7

Kihl: 2.1-2.3, 3.1, 3.5

Data and signals

Data = The information we want to transmit.

Signal = The representation of data when it is transmitted on a link.

Data

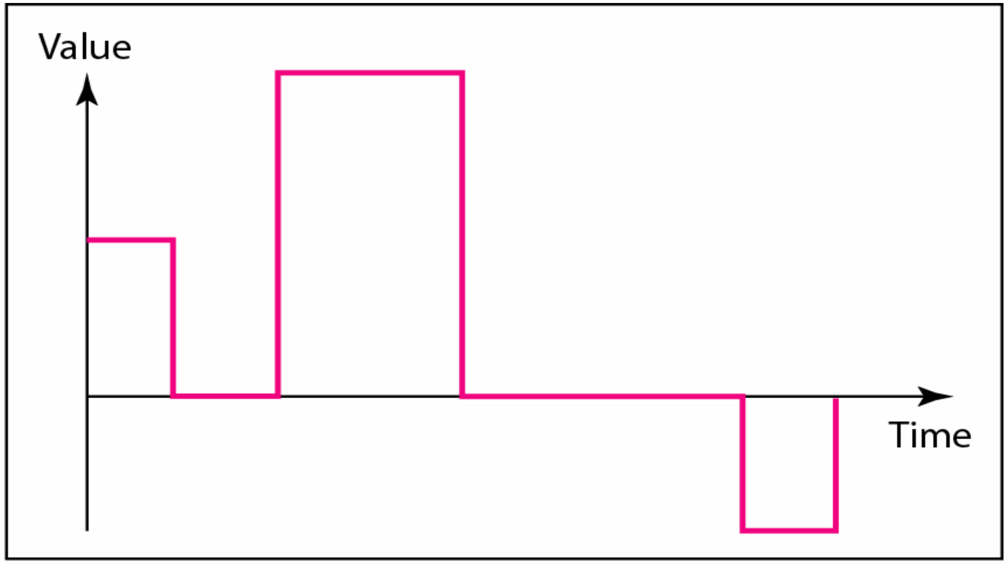
- Analog data: Information that is continuous.
  - Examples: Audio, Images
- Digital data: Information that has discrete states.
  - Example: Text
- When we need analog data in a digital format, a *digitization process* is needed.

Signals

- Analog signals have infinitely many levels of intensity over a period of time.
    - Composite sine waves.
- Digital signals have only a limited number of defined values.
    - Different levels of voltage amplitudes.

Analog v. Digital signals

a. Analog signal



b. Digital signal

Note

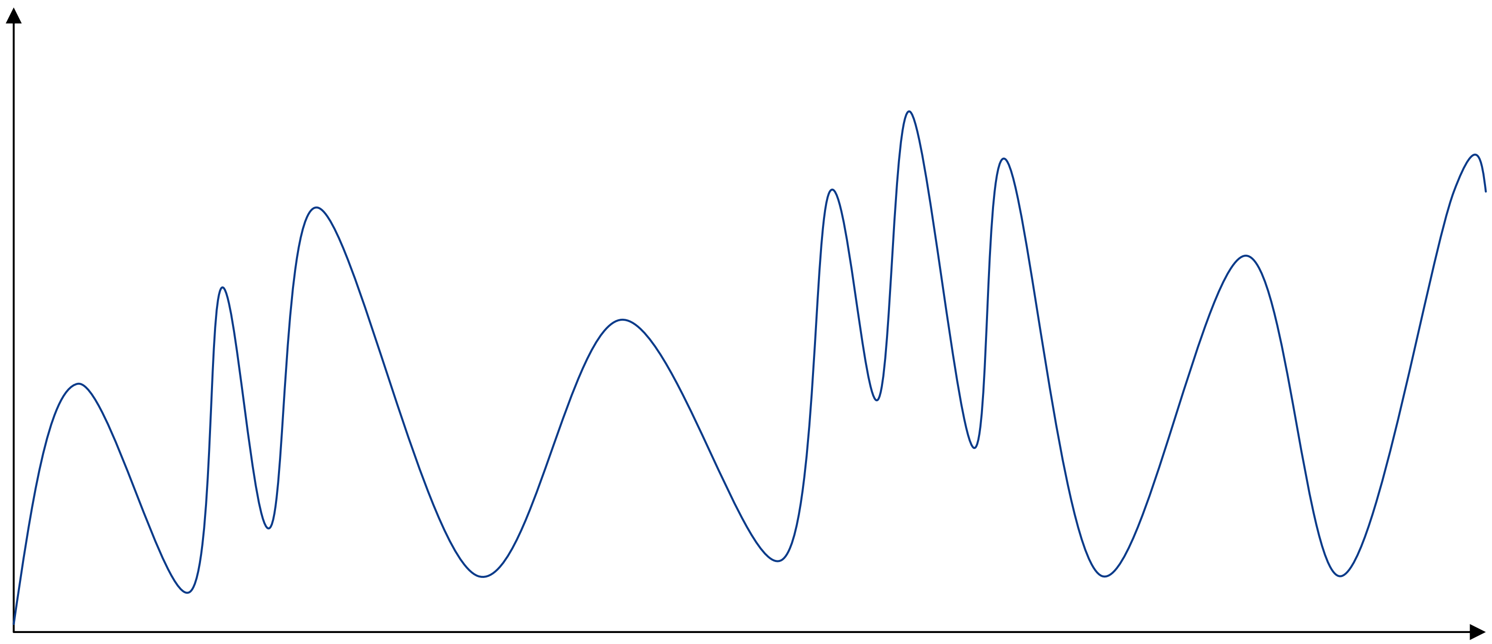
According to Fourier analysis, any composite signal is a combination of simple sine waves with different frequencies, amplitudes, and phases.

All combinations are possible…

- Digital data – Digital signal
  - Ethernet
- Analog data – Digital signal
  - Telephone transport network
- Digital data – Analog signal
  - WLAN, 3G, GSM, ADSL, Digital radio
- Analog data – Analog signal
  - FM radio, TV, Telephone access network

Digitization

Analog data has has infinite levels of intensity.



It must be discretized!

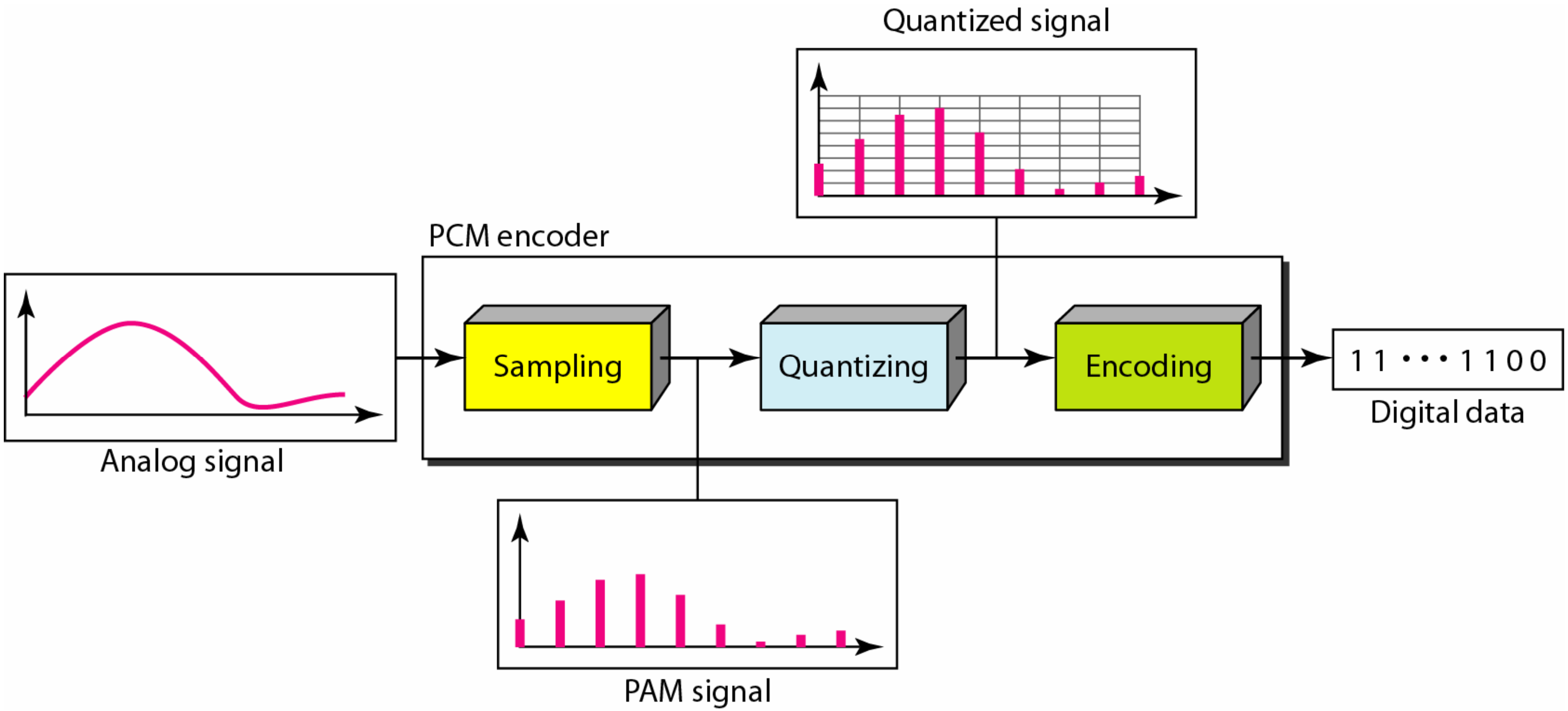
Digitization of analog data

The process is performed in three steps:

1. Sampling
2. Quantization
3. Encoding

This simplest version of this digitization process is called *Pulse Code Modulation (PCM).*

Pulse Code Modulation



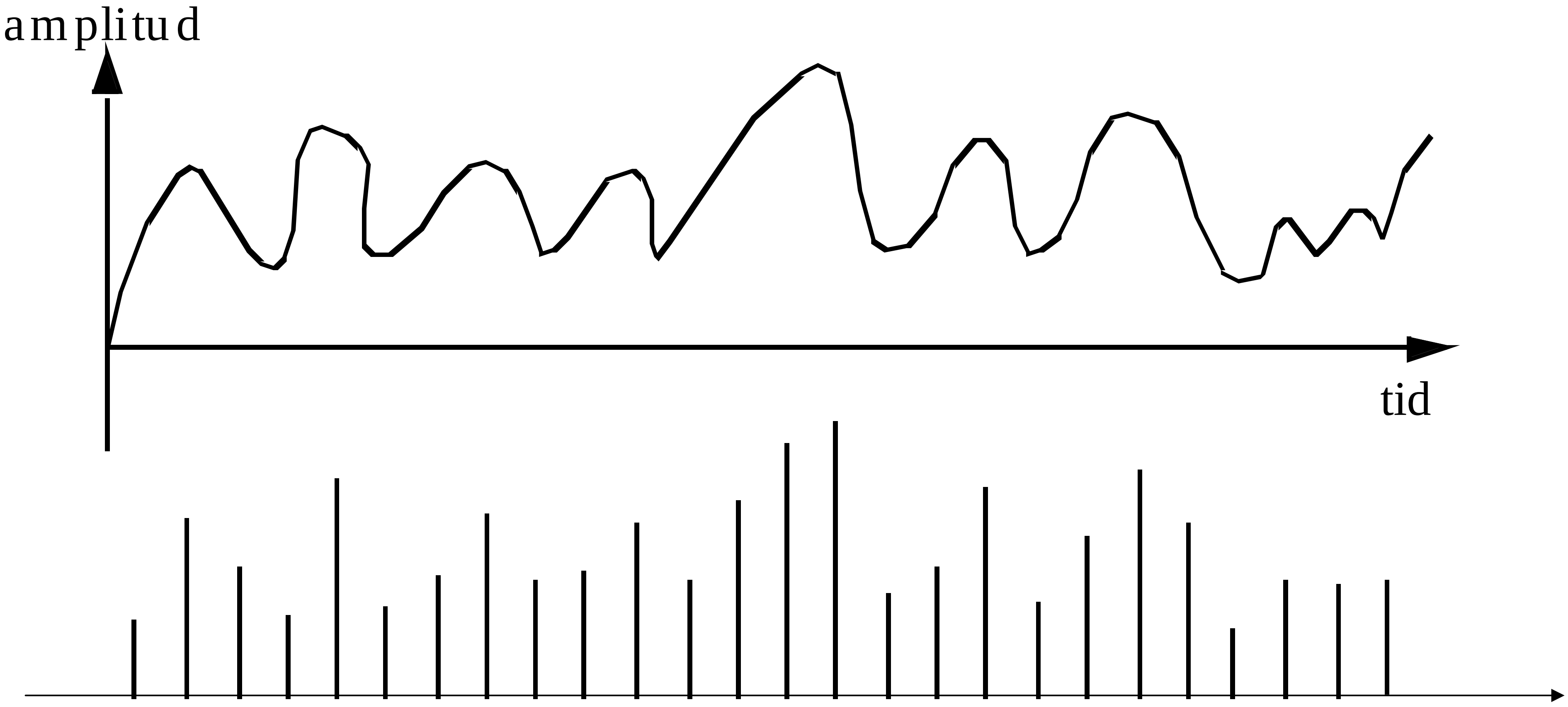
Sampling (1)

The analog signal is sampled with a specified *sampling rate*.

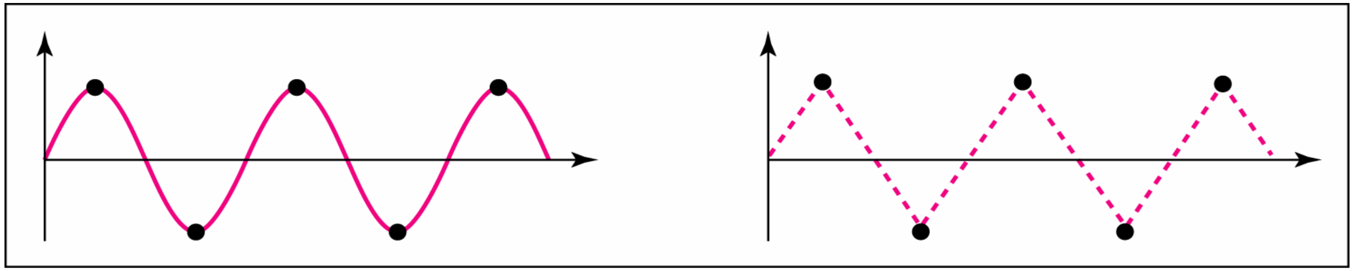
An analog signal is composed of a number of frequencies.

If the highest frequency is N Hz, the signal must be sampled with the frequency $2N$ Hz. Otherwise, it cannot be reproduced.

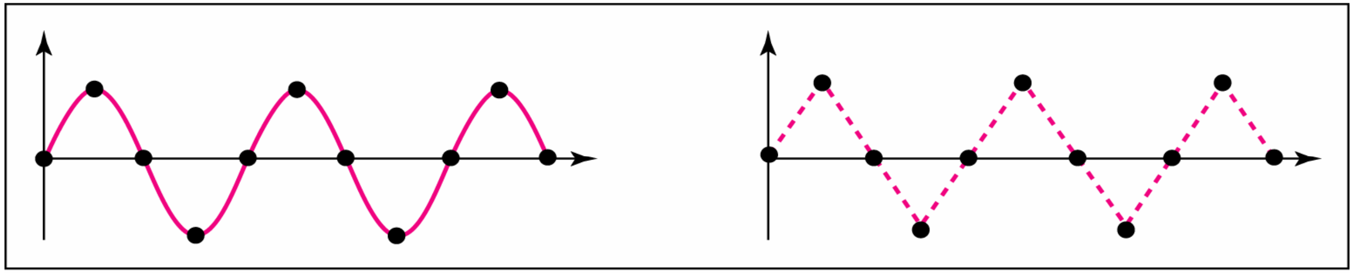
Sampling (2)



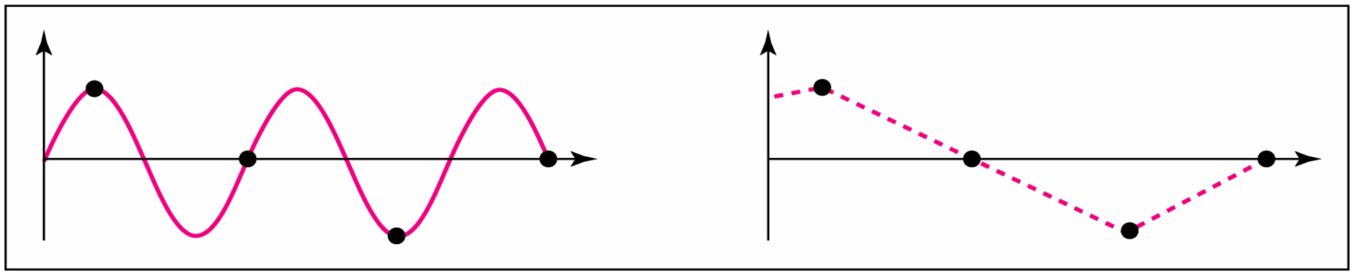
Example of sampling rates



a. Nyquist rate sampling: $f_s = 2\,f$



b. Oversampling: $f_s = 4\,f$



c. Undersampling: $f_s = f$

Quantization

The sampled values must be quantized to a finite number of amplitude levels.

The number of amplitude levels determine the number of bits that are required to represent the signal in the encoding process.

Example: 256 levels require 8 bits ($2^8=256$).

How many signal levels?

The number of amplitude levels determine the quality of the signal when it is reproduced. There will always be a *quantization error*.

Telephony: 8 bits = 256 levels.

CD: 16 bits = 65.536 levels.

Encoding

Each amplitude level is represented by a bit sequence.

The digital data can then be stored, and later recovered with a PCM decoder.

Data transfer on a link

Two computers communicate on a link.

The physical link is called the **Transmission Medium**.

Transmission Medium

”Anything that can carry information from a source to a destination.”

- Guided media (wired communication):
  - Twisted-pair cable, Coaxial cable, Fiber-optic cable
- Unguided media (wireless communication):
  - Radio, Microwaves, Infrared

Bandwidth

- Analog definition: The range of frequencies a link can pass (measured in *Hz*).
- Digital definition: The number of bits per second that a link can transmit (measured in *bps*). Also called *bit rate*.

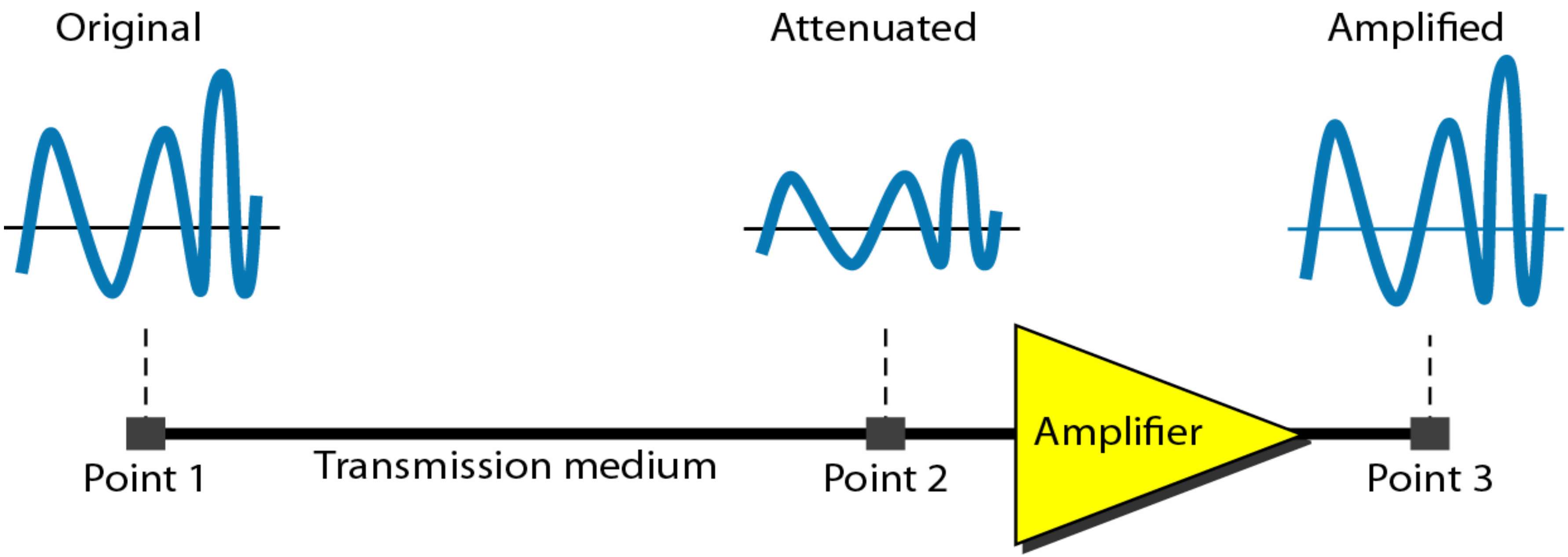
Transmission impairment

When a signal travels on a link, it will deteriorate due to transmission impairment.

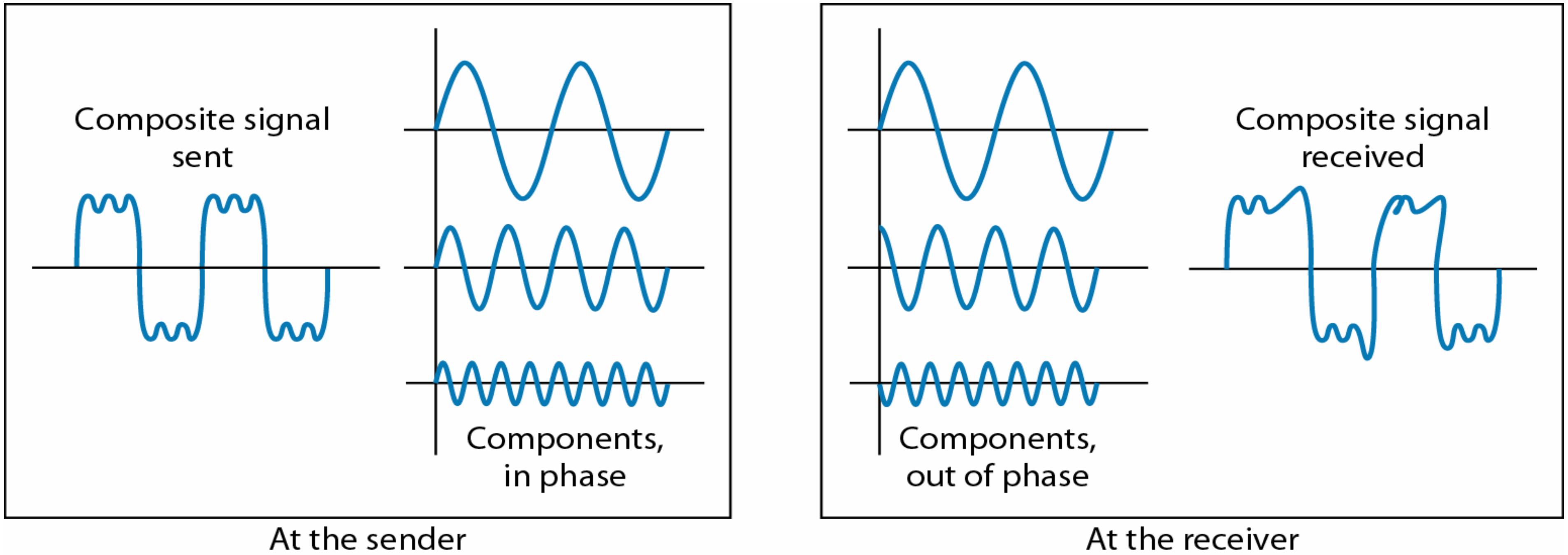
- Attenuation (swe: dämpning): Loss of energy
- Distortion: Change of signal shape
- Noise: The signal is corrupted due to, e.g., thermal noise or crosstalk (swe: överhörning).

$$\text{Signal-to-noise ratio (SNR)} = \frac{\text{Average signal power}}{\text{Average noise power}}$$

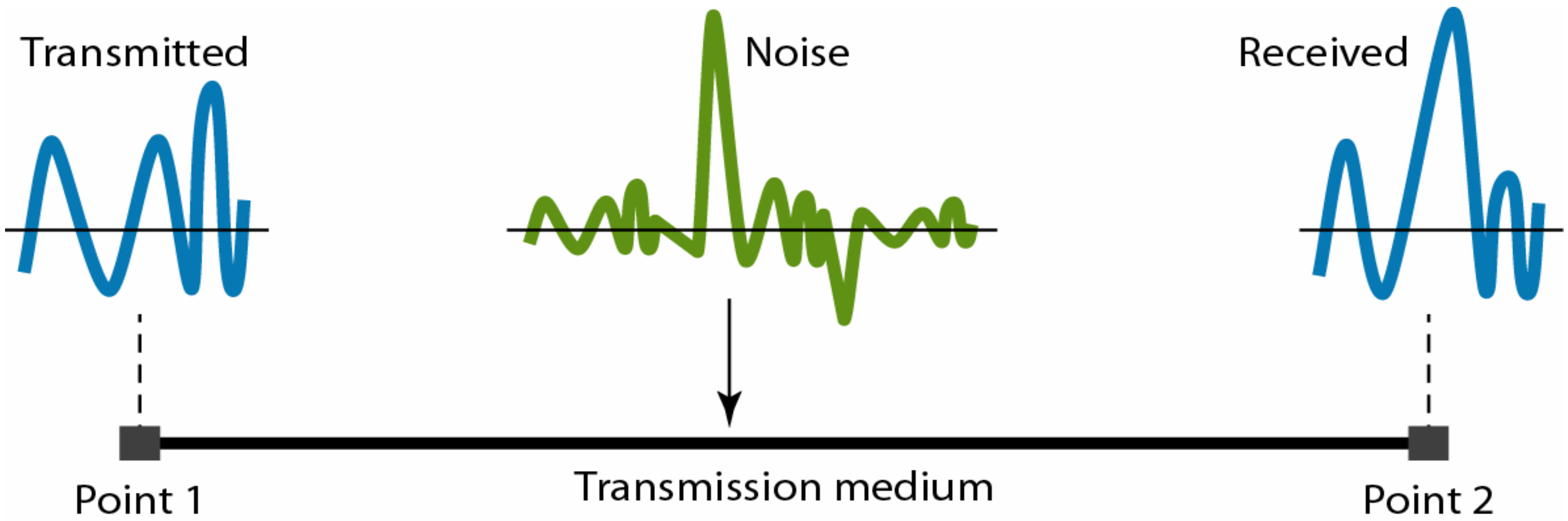
Attenuation



Distortion


Composite signal
sent
Components,
in phase
At the sender
Components,
out of phase
Composite signal
received
At the receiver

Noise


Transmitted
Noise
Received
Point 1
Transmission medium
Point 2

Data rate limits

- Nyquist bit rate
  - The theoretical maximum bit rate for a noiseless channel.
  - Bit rate = 2 x Bandwidth x $\log_2 L$
- Shannon capacity
  - The theoretical highest bit rate for a noisy channel
  - Bit rate = Bandwidth x $\log_2 (1+SNR)$

Example: Bit rate for telephony links

- Analog signal in frequencies 0 - 4kHz.
- Nyquist theorem gives that the sampling frequency is 2x4 kHz = 8000 samples per second.
- 8-bit encoding.

The maximum bit rate is 64 kbps.

Performance metrics

- Throughput: The "real" transmission capacity between a source and a destination (bits/sec).
- Latency: The time required to send a message from a source to a destination. Is a sum of *propagation time*, *transmission time*, *queuing time* and *processing delay*.
- Bandwidth-delay product:
  Bandwidth (Throughput) x Latency

Example: Bandwidth-delay product